3/09/26

EXPERIMENT-10 A

INTERNETWORKING USING ROUTERS

AIM:

To design & configure a simple internetwork using a router & 2 in CISCO packet tracer & test connectivity using PPU.

PROCEDURE:

STEP 1: Configure Router 1

1) Open cisco packet tracer
2) select router → Router 1
3) Open CLI & press enter
4) Configure FastEthernet 0:

   Interface FastEthernet 0/0
   IP address: 192.168.10.12.255.255.255.0
   no shut down

5) Configure FastEthernet 0:

   Interface FastEthernet 0/1
   IP address: 192.168.101.255.255.255.0
   No shut down

STEP 2: Configure Pc0:

   go to Pc0 → desktop → IP configuration
   IP address: 192.168.10.2
   Subnet Host: 255.255.255.0
   Default gateway: 192.168.10.1
   Same for PC1.

STEP 3: Connect the devices

Use the copper straight-through cables

PC0 Fast Ethernet 0 → Router 1 fast Ethernet 0/0

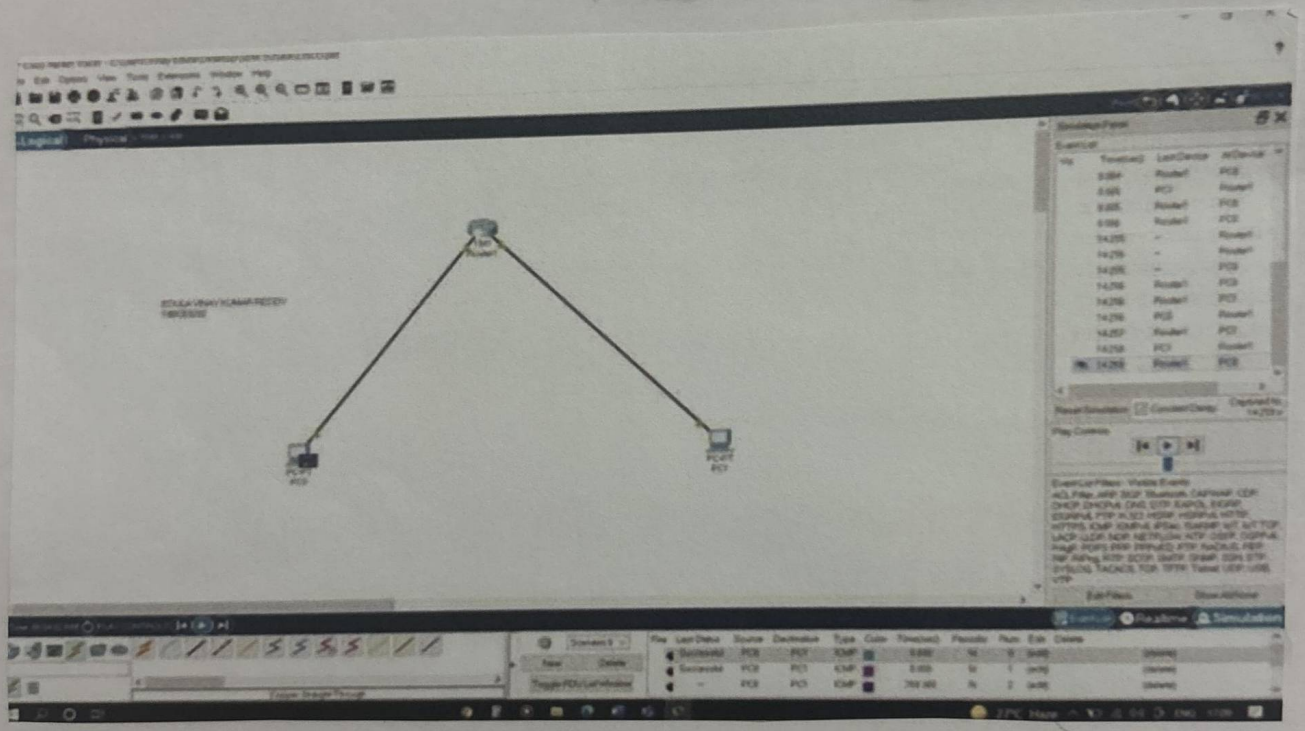
PC1 Fast Ethernet 0 → Router 1 fast Ethernet 0/1

STEP 4: Test connectivity

1) Select simulation mode
2) Choose add simple PDU
3) Click PC0 → PC1
4) If the PDU is successful, the 2 network are communicating through the router.

RESULT:

The internetwork was successfully designed & configured & connectivity between PC0 & PC1 was verified.



EXPERIMENT - 10B

WIRELESS ROUTER, DHCP SERVER AND INTERNET CLOUD

AIM:

To design & configure an internetwork using a wireless router, DHCP server & Internet & verify connectivity.

PROCEDURE:

STEP 1: Build the Network

Add:

→ PC
→ Wireless Router
→ Laptop
→ cable Modem
→ Internet Cloud
→ cisco.com server

STEP 2: Connect Devices

Use the appropriate cables

PC → Wireless Router → cable Modem:
copper straight-through

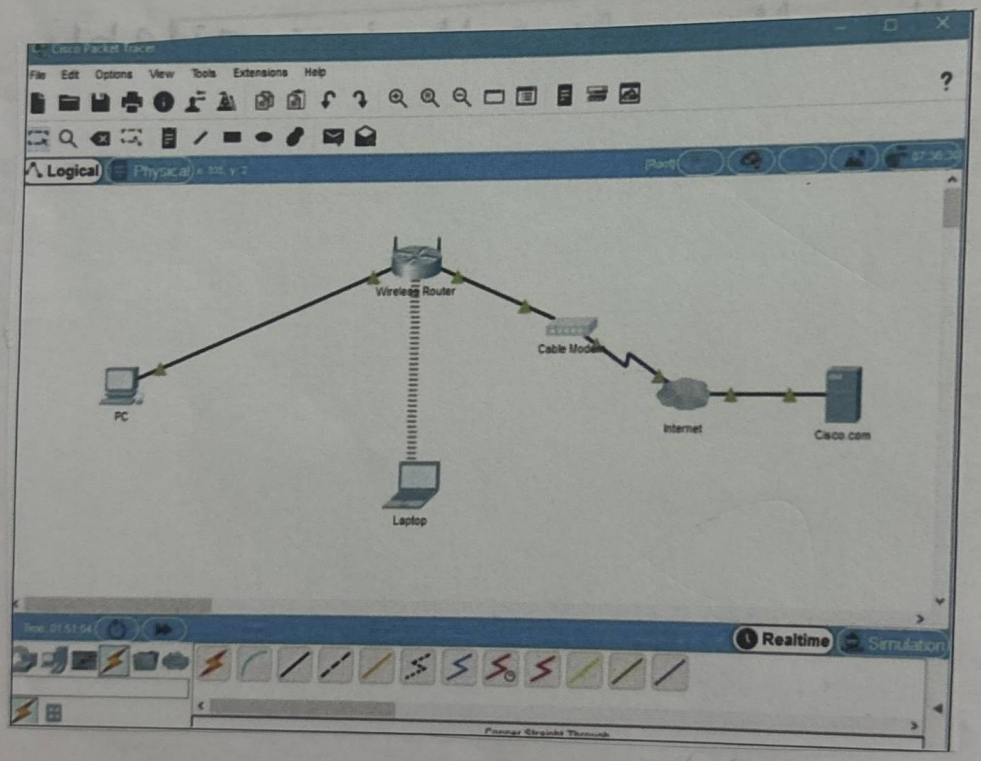
* Wireless Router → cable Modem:
copper straight-through

* cable modem → Internet cloud:
coaxial cable

* Internet cloud → ciscocom server:
copper straight-through

STEP 3: configure wireless Router

(1) Open wireless Router → GUI
(2) Go to wireless
(3) Set DNS server as 208.67.220.200.220
(4) Click save settings


Cisco Packet Tracer
File Edit Options View Tools Extensions Help
Logical
Physical
Wireless Router
Cable Modem
PC
Internet
Cisco.com
Laptop
Realtime
Simulation

STEP 4: Configure cisco.com server

DHCP

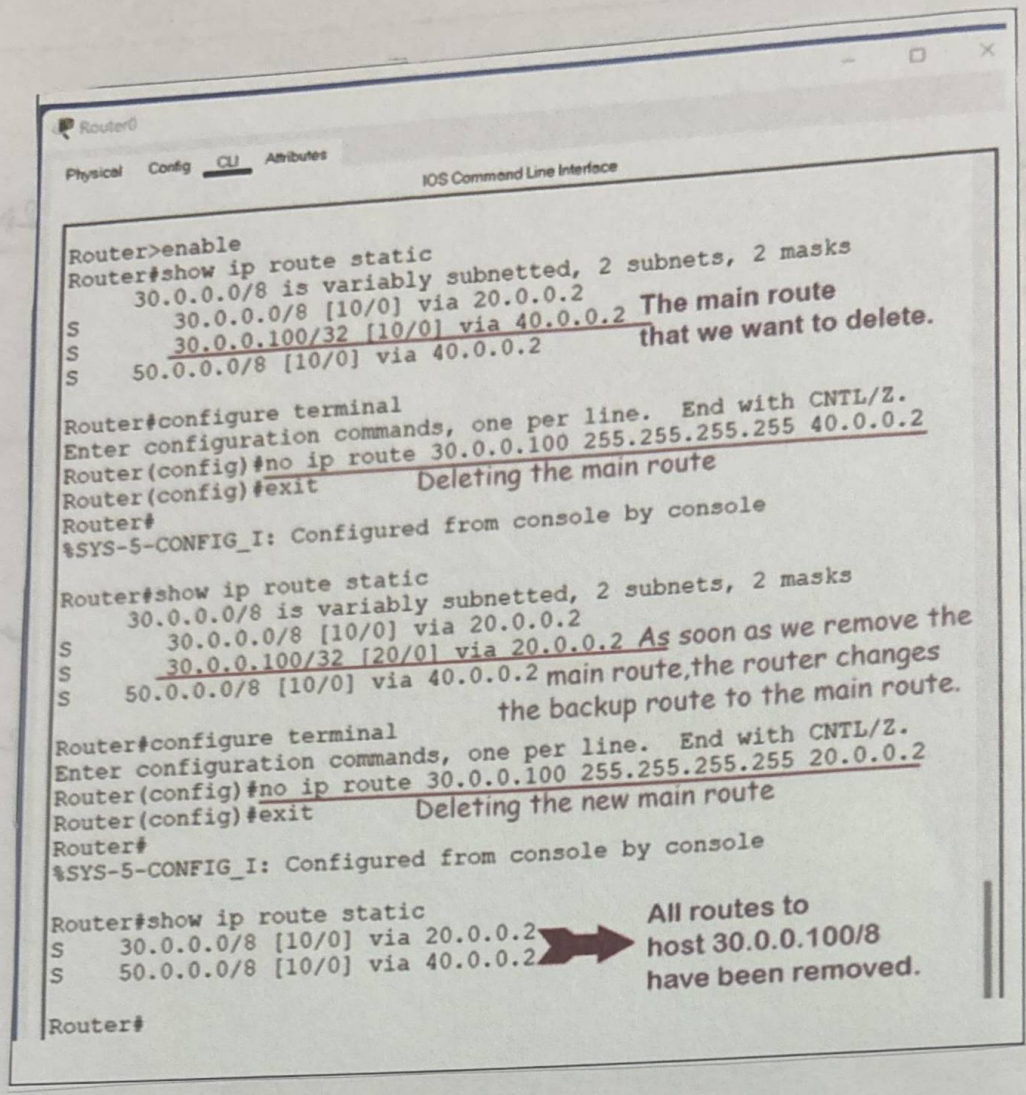
* DHCP : On
* pool name : DHCP pool
* default gateway : 208.67.220.220
* Starting IP : 208.67.220.1
* Maximum Users : 50
* click add

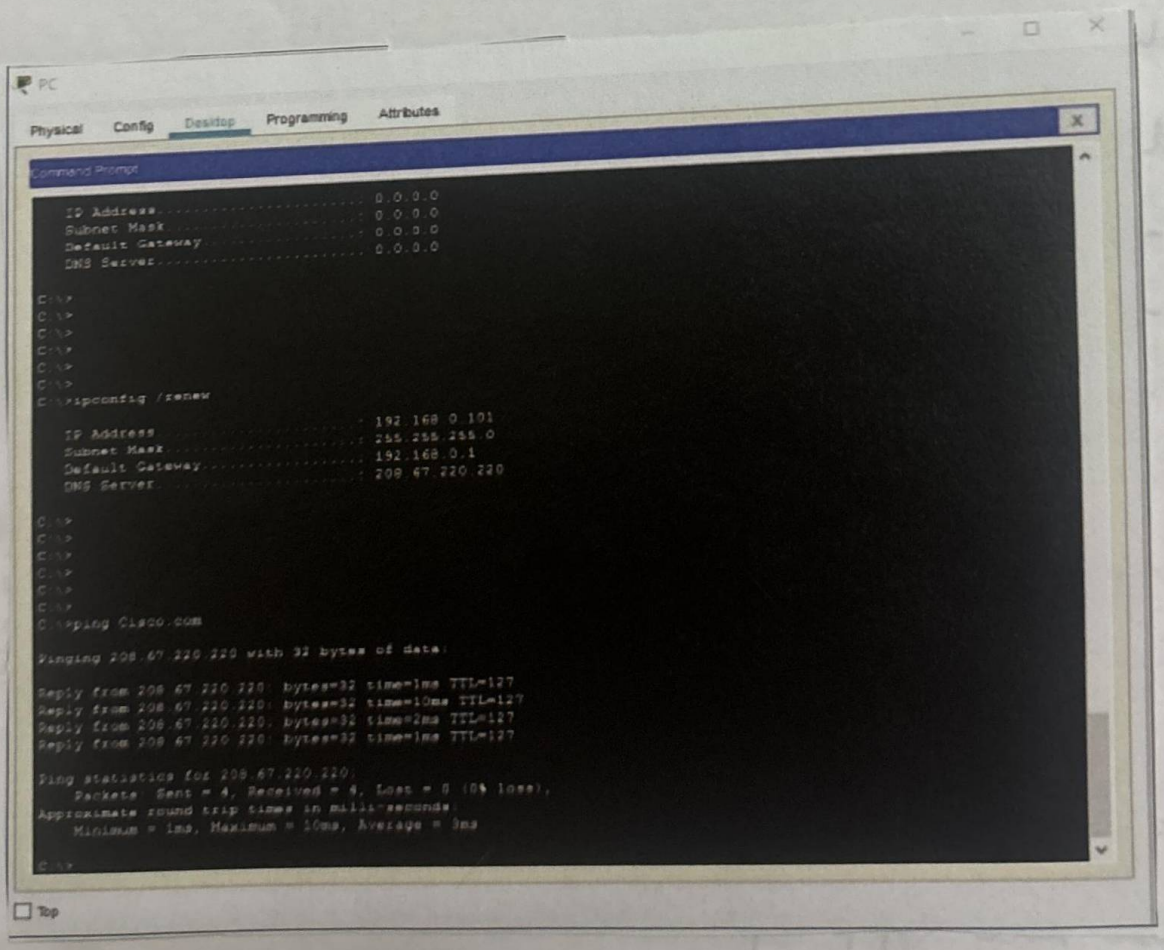
STEP 5: Test connectivity:

(1) Open pc → Desktop → command prompt
(2) Type : ipconfig/release
(3) Then : ipconfig/renew
(4) test the server : ping cisco.com

RESULT:

The wireless internetwork was successfully designed using a wireless router, DHCP server & internet cloud & connectivity was verified using the ping command.


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>enable
Router#show ip route static
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S 30.0.0.0/8 [10/0] via 20.0.0.2
S 30.0.0.100/32 [10/0] via 40.0.0.2
S 50.0.0.0/8 [10/0] via 40.0.0.2
The main route that we want to delete.
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip route 30.0.0.100 255.255.255.255 40.0.0.2
Router(config)#exit
Deleting the main route
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip route static
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S 30.0.0.0/8 [10/0] via 20.0.0.2
S 30.0.0.100/32 [20/0] via 20.0.0.2
S 50.0.0.0/8 [10/0] via 40.0.0.2
As soon as we remove the main route, the router changes the backup route to the main route.
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip route 30.0.0.100 255.255.255.255 20.0.0.2
Router(config)#exit
Deleting the new main route
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip route static
S 30.0.0.0/8 [10/0] via 20.0.0.2
S 50.0.0.0/8 [10/0] via 40.0.0.2
All routes to host 30.0.0.100/8 have been removed.
Router#


PC
Physical
Config
Desktop
Programming
Attributes
Command Prompt
IP Address..... 0.0.0.0
Subnet Mask..... 0.0.0.0
Default Gateway..... 0.0.0.0
DNS Server..... 0.0.0.0
C:\>ipconfig /renew
IP Address..... 192.168.0.101
Subnet Mask..... 255.255.255.0
Default Gateway..... 192.168.0.1
DNS Server..... 208.67.220.220
C:\>ping Cisco.com
Pinging 208.67.220.220 with 32 bytes of data:
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=10ms TTL=127
Reply from 208.67.220.220: bytes=32 time=2ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Ping statistics for 208.67.220.220:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 10ms, Average = 3ms
C:\>
Top